

A neural network-based fuzzy time series model to improve forecasting

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ABSTRACT

Neural networks have been popular due to their capabilities in handling nonlinear relationships. Hence, this study intends to apply neural networks to implement a new fuzzy time series model to improve forecasting. Differing from previous studies, this study includes the various degrees of membership in establishing fuzzy relationships, which assist in capturing the relationships more properly. These fuzzy relationships are then used to forecast the stock index in Taiwan. With more information, the forecasting is expected to improve, too. In addition, due to the greater amount of information covered, the proposed model can be used to forecast directly regardless of whether out-of-sample observations appear in the in-sample observations. This study performs out-of-sample forecasting and the results are compared with those of previous studies to demonstrate the performance of the proposed model.

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1. Introduction

Fuzzy time series models, a counterpart of conventional time series models, have become increasingly popular in recent years. Various fuzzy time series models have been proposed, including first-order models (Chen, 1996; Chen & Hwang, 2000; Huarng & Yu, 2005, 2006b; Hwang, Chen, & Lee, 1998; Lee & Chou, 2004; Song & Chissom, 1993a, 1994; Sullivan & Woodall, 1994; Tseng, Tzeng, & Yu, 1999; Yu, 2005a, 2005b), high-order models (Chen, 2002; Huarng & Yu, 2003a; Li & Cheng, 2007; Nguyen & Wu, 2000), seasonal models (Chang, 1997; Song, 1999; Tseng et al., 1999), bivariate models (Chu, Chen, Cheng, & Huang, 2009; Egrioglu, Aladag, Yolcu, Basaran, & Uslu, in press; Hsu, Tse, & Wu, 2003; Yu & Huarng, 2008), multivariate models (Cheng, Cheng, & Wang, 2008; Huarng, 2001a; Huarng, Yu, & Hsu, 2007; Jilani & Burney, 2007, 2008; Teoh, Chen, Cheng, & Chu, in press; Wu & Hsu, 2002), and hybrid models (Huarng & Yu, 2003b; Jilani & Burney, 2007; Jilani, Burney, & Ardil, 2007; Lee, Wang, Chen, & Leu, 2006; Own & Yu, 2005; Tseng & Tzeng, 2002). A number of other studies have also focused on the partitioning of fuzzy sets to improve forecasting results (Huarng, 2001a, 2001b; Huarng & Yu, 2004, 2006a, 2006b; Jilani, Burney, & Ardil, 2008; Yu, 2005a, 2005b).

These studies have been applied to various problem domains, such as temperature (Chen & Hwang, 2000; Lee et al., 2006; Lee, Wang, & Chen, 2007, 2008; Wang & Chen, 2009), enrollment (Chen, 1996, 2002; Huarng, 2001a, 2001b; Hwang et al., 1998; Jilani et al.,

2008; Lee & Chou, 2004; Nguyen & Wu, 2000; Song & Chissom, 1993a; Song & Chissom, 1994; Sullivan & Woodall, 1994; Wang, 2004), financial indices (Chen, Cheng, & Teoh, 2007; Hsu et al., 2003; Huarng, 2001a, 2001b; Huarng & Yu, 2003a, 2003b; Huarng, Yu, et al., 2007; Lee et al., 2006; Nguyen & Wu, 2000; Yu, 2005a, 2005b; Lee et al., 2007, 2008; Wang & Chen, 2009; Yu & Huarng, 2008; Cheng, Chen, Teoh, & Chiang, 2008; Cheng & Wei, in press), tourism demand (Huarng, Moutinho, & Yu, 2007; Wang, 2004; Wang & Hsu, 2008), and car road accidents (Jilani & Burney, 2007; Jilani et al., 2007). Many of these models have been shown to outperform their more conventional counterpart models (Chen, 1996; Huarng, Moutinho, et al., 2007; Hwang et al., 1998; Song & Chissom, 1993a, 1994).

The application of these fuzzy time series models mainly consists of three major steps: fuzzification (the partitioning of fuzzy sets and fuzzifying observations), establishing fuzzy relations (different methods), and defuzzification. Although various studies may place emphasis on different steps, most studies focus on establishing fuzzy relations. However, because the fuzzy relationships can be nonlinear and complex, a powerful method is thus needed to calculate these relationships.

Meanwhile, neural networks have recently attracted more attention in forecasting (Donaldson & Kamstra, 1996; Kanas, 2001; Law, 2000; Tugba & Casey, 2005), and they have been popular in terms of their ability to handle nonlinear problems. Hence, this study believes that neural networks may be appropriate for computing the fuzzy relationships in fuzzy time series.

This study intends to propose a neural network fuzzy time series model, where in-sample observations are used for training and out-of-sample observations for forecasting. The challenge of conducting out-of-sample forecasting is that there may be some

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out-of-sample observations that never appear in the in-sample observations.

The contribution of this study is, first, that the proposed model applies neural networks to better capture the fuzzy relationships and then to forecast better. Second, all the degrees of membership will be used in training as well as forecasting. Because it covers more information (in contrast to some other studies), the proposed model is expected to forecast better, too. Third, because it considers more information, the proposed model can be used to forecast directly, even though some out-of-sample observations may not appear in the in-sample observations.

The objective of this study is to capture the fuzzy relationships more properly and thereby improve the fuzzy time series forecasting. To that end, the remainder of this paper consists of the following sections. Section 2 reviews the concepts of fuzzy time series and neural networks. Section 3 describes the data. Section 4 explains the proposed model. Section 5 uses an example to demonstrate the forecasting process. Section 6 compares the empirical results. Section 7 concludes the paper.

2. Literature review

2.1. Fuzzy time series model

Conventional time series refer to real numbers, but fuzzy time series are structured by fuzzy sets (Chen, 1996). Let U be the universe of discourse, such that $U = \{u_1, u_2, \dots, u_n\}$. A fuzzy set A of U is defined as $A = f_A(u_1)/u_1 + f_A(u_2)/u_2 + \dots + f_A(u_n)/u_n$, where f_A is the membership function of A , and $f_A: U \rightarrow [0, 1]$. $f_A(u_i)$ is the grade of membership of u_i in A , where $f_A(u_i) \in [0, 1]$ and $1 \leq i \leq n$.

Definition 1. Let $Y(t)$ ($t = \dots, 0, 1, 2, \dots$), a subset of a real number, be the universe of discourse on which fuzzy sets $f_i(t)$ ($i = 1, 2, \dots$) are defined and $F(t)$ is a collection of $f_1(t), f_2(t), \dots, F(t)$ is referred to as a fuzzy time series on $Y(t)$. Here, $F(t)$ is viewed as a linguistic variable and $f_i(t)$ can represent possible linguistic values of $F(t)$ (Song & Chissom, 1993b).

If $F(t)$ is caused by $F(t-1)$ only, the relationship can be expressed as $F(t-1) \rightarrow F(t)$ (Chen, 1996). Various operations have been applied to compute the fuzzy relationship between $F(t)$ and $F(t-1)$. These operations could be as complicated as matrix multiplications (Hwang et al., 1998; Song & Chissom, 1994; Sullivan & Woodall, 1994).

Chen (1996) suggested that when the maximum degree of membership of $F(t)$ belongs to A_i , $F(t)$ is considered to be A_i . Hence, $F(t-1) \rightarrow F(t)$ becomes $A_i \rightarrow A_j$. Only the most significant degrees of membership were taken into consideration in his study. From then on, many models were proposed by following this concept (Huarng, 2001a, 2001b; Huarng & Yu, 2003a, 2003b, 2005, 2006a, 2006b). The major problem that these studies faced is that the out-of-sample observations may not appear in the in-sample observations. In this case, there is no proper $A_i \rightarrow A_j$ that can be applied. *Ad-hoc* methods are thus needed to solve this problem. In this study, this problem will be solved using the proposed model.

2.2. Neural network models

Neural networks mimic human intelligence in deducing or learning from observations, and usually consist of an input layer, an output layer, and one or more hidden layers. Each of the layers contains nodes, and the nodes of two consecutive layers are connected with each other. Neural networks are able to discover complex nonlinear relationships in the observations (Donaldson & Kamstra, 1996; Indro, Jiang, Patuwo, & Zhang, 1999). Some applications of neural networks include credit ratings (Kumar & Bhattacharya, 2006), Dow Jones forecasting (Kanas, 2001), customer

satisfaction analysis (Gronholdt & Martensen, 2005), stock ranking (Refenes, Azema-Barac, & Zaprani, 1993), and tourism demand (Law, 2000; Law & Au, 1999; Martin & Witt, 1989; Palmer, Montañó, & Sesé, 2006).

One of the fuzzy time series models has successfully applied neural networks to improve forecasting results (Huarng & Yu, 2006b). However, this particular model has only used the most significant degrees of membership for each observation for training and forecasting. The other degrees of membership were ignored, and this may affect the forecasting.

3. Data description

This study uses daily close prices of the stock index in Taiwan (the Taiwan Stock Exchange Capitalization Weighted Stock Index or TAIEX) as the forecasting target. The observations¹ extend from 2000 to 2004. Since some studies have mentioned that estimation alone cannot be claimed for good forecasting (Martin & Witt, 1989), this study measures performance based on the out-of-sample forecasting. For each year, the in-sample observations, which are used for neural network training, cover the period from January to September, and the out-of-sample observations, which are used for neural network forecasting, cover the period from October to December.

4. The theoretical model

A theoretical model is proposed, for which the various steps are explained as follows:

Step 1. Difference

Many previous studies conducted fuzzy time forecasting on the observations directly (Chen, 1996, 2002; Huarng, 2001a, 2001b; Hwang et al., 1998; Song & Chissom, 1993a, 1994). This study forecasts the differences in the observations instead, by obtaining the differences between every two consecutive observations at t and $t-1$:

$$d(t-1, t) = \text{obs}(t) - \text{obs}(t-1), \quad (1)$$

where $\text{obs}(t)$ and $\text{obs}(t-1)$ are two consecutive observations at t and $t-1$; $d(t-1, t)$ is their difference.

Step 2. Adjustment

The differences may turn out to be negative. To ensure that all the universes of discourse are positive, we add different positive constants to the differences for different years:

$$d'(t-1, t) = d(t-1, t) + \text{const}. \quad (2)$$

For each year, we obtain the minimum and maximum of all the differences, $D_{\min}$ and $D_{\max}$:

$$\begin{aligned} D_{\min} &= \min(d'(t-1, t)), \quad \text{for all } t, \\ D_{\max} &= \max(d'(t-1, t)), \quad \text{for all } t. \end{aligned} \quad (3)$$

Step 3. Universe of Discourse

Following (Chen, 1996), the universe of discourse, U , can be defined as $[D_{\min} - D_1, D_{\max} + D_2]$, where D_1 and D_2 are two proper positive numbers. Suppose the length of the interval is set to l . We then separate U into equal intervals and name them $u_1, u_2, u_3, \dots$, where

$$\begin{aligned} u_1 &= [D_{\min} - D_1, D_{\min} - D_1 + l], \\ u_2 &= [D_{\min} - D_1 + l, D_{\min} - D_1 + 2 \times l], \\ &\dots \\ u_k &= [D_{\min} - D_1 + (k-1) \times l, D_{\min} - D_1 + k \times l]. \end{aligned} \quad (4)$$

¹ The data are from the TEJ.

Their corresponding midpoints are

$$\begin{aligned} m^1 &= \frac{D_{\min} - D_1 + D_{\min} - D_1 + l}{2} = D_{\min} - D_1 + \frac{l}{2}, \\ m^2 &= \frac{D_{\min} - D_1 + l + D_{\min} - D_1 + 2 \times l}{2} = D_{\min} - D_1 + \frac{3 \times l}{2}, \\ &\dots \\ m^k &= D_{\min} - D_1 + \frac{(2 \times k - 1) \times l}{2}. \end{aligned} \quad (5)$$

Define the linguistic values of the fuzzy sets. Let $A_1, A_2, A_3, \dots$ be linguistic values, and label all the fuzzy sets by possible linguistic values, $u_1, u_2, u_3, \dots$

Step 4. Fuzzification

Next, $d'(t-1, t)$ can be fuzzified into a set of degrees of membership, $V(t-1, t)$, where

$$V(t-1, t) = [\mu_{t-1,t}^1, \mu_{t-1,t}^2, \dots]. \quad (6)$$

Step 5. Neural Network Training

To improve forecasting results, this study uses all the degrees of membership to establish fuzzy relationships. Two consecutive V 's can be used to establish a fuzzy relationship:

$$V(t-1, t) \rightarrow V(t, t+1). \quad (7)$$

This study then applies back-propagation neural networks to establish (or train) the fuzzy relationships. The neural network structure consists of one input layer, one hidden layer, and one output layer. The numbers of input and output nodes are equal to the number of degrees of membership in $V(t-1, t)$. The number of hidden nodes is set to the sum of the number of input and output nodes. The neural network structure is shown in Fig. 1.

Step 6. Neural Network Forecasting

With $V(t-1, t)$, we can proceed to forecast $V(t, t+1)$ by means of the trained neural network. If we use the in-sample observations, the results are referred to as those of the in-sample estimation. If we use out-of-sample observations, the results are referred to as those of out-of-sample forecasting.

Step 7. Defuzzification

This study applies weighted averages to defuzzify the degrees of membership:

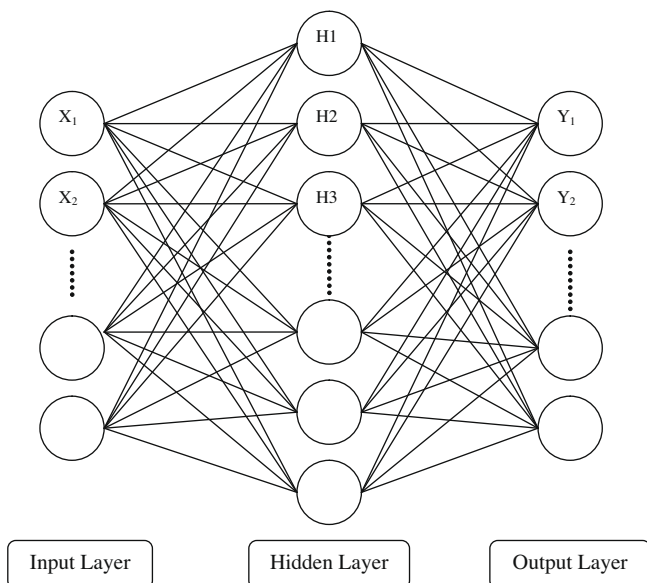


Fig. 1. A neural network structure.

$$fd(t-1, t) = \frac{\sum_{k=1}^n \mu_{t-1,t}^k \times m^k}{\sum_{k=1}^n \mu_{t-1,t}^k}, \quad (8)$$

where $fd(t-1, t)$ is the forecasted difference between $t-1$ and t ; $\mu_{t-1,t}^k$ represents the forecasted degrees of membership and m^k denotes the corresponding midpoints of the interval, $\mu_{t-1,t}^k$.

Step 8. Forecasting

Once we obtain the forecasted difference between $t-1$ and t , we can calculate the forecast for t as follows:

$$fd'(t-1, t) = fd(t-1, t) - \text{const}, \quad (9)$$

$$\text{forecast}(t) = fd'(t-1, t) + \text{obs}_{t-1}. \quad (10)$$

Step 9. Performance Evaluation

We follow previous studies (Huang & Yu, 2006a, 2006b) by using the root mean squared error (RMSE) to conduct the performance evaluation. The RMSE is calculated as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{t=k+1}^n (\text{forecast}(t) - \text{obs}(t))^2}{n-k}}, \quad (11)$$

where there are n observations, including k in-sample and $n-k$ out-of-sample observations.

5. A forecasting example

The proposed model is applied to forecast the stock index, the TAIEEX. We take the year 2000 to illustrate each step of the model.

Step 1. Difference

The stock index for 2000/1/5 is 8850 and that for 2000/1/4 is 8757. Hence,

$$d(2000/1/4, 2000/1/5) = 93.$$

Step 2. Adjustment

For the year 2000, the minimum of all the differences is -618 . Hence, 700 is considered to be appropriate as the constant for the year 2000:

$$\text{const} = 700.$$

The difference in the previous step is adjusted as follows:

$$d'(2000/1/4, 2000/1/5) = d(2000/1/4, 2000/1/5) + \text{const} = 793.$$

Then, for each year, we can get the minimum and maximum of all the differences, $D_{\min}$ and $D_{\max}$. For the year 2000, $D_{\min} = 82$, $D_{\max} = 1168$.

Step 3. Universe of Discourse

Following Chen (1996), the universe of discourse, U , can be defined as $[D_{\min} - D_1, D_{\max} + D_2]$, where D_1 and D_2 are two proper positive numbers. The purpose of D_1 and D_2 is to make the lower and upper bounds of U become multiples of hundreds and thousands, etc. For the year 2000, $D_{\min} = 82$ and $D_{\max} = 1168$. Hence, $D_1 = 82$ and $D_2 = 32$. Hence, $U = [0, 1200]$.

Many studies set the length of the interval according to their study domains (Chen, 1996; Chen & Hwang, 2000; Huang, 2001a; Huang, Moutinho, et al., 2007; Huang & Yu, 2006b). Following these studies, the length of the interval is set to 100. We then separate U into equal intervals and name them $u_1, u_2, u_3, \dots, u_{12}$, where $u_1 = [0, 100]$, $u_2 = [100, 200]$, $u_3 = [200, 300]$, $\dots$, $u_{12} = [1100, 1200]$. Let $A_1, A_2, A_3, \dots, A_{12}$ be linguistic values, and label all the fuzzy sets by possible linguistic values, $u_1, u_2, u_3, \dots, u_{12}$.

Step 4. Fuzzification

In Fig. 2, $d'(2000/1/4, 2000/1/5) = 793$ can be fuzzified into the following degrees of membership:

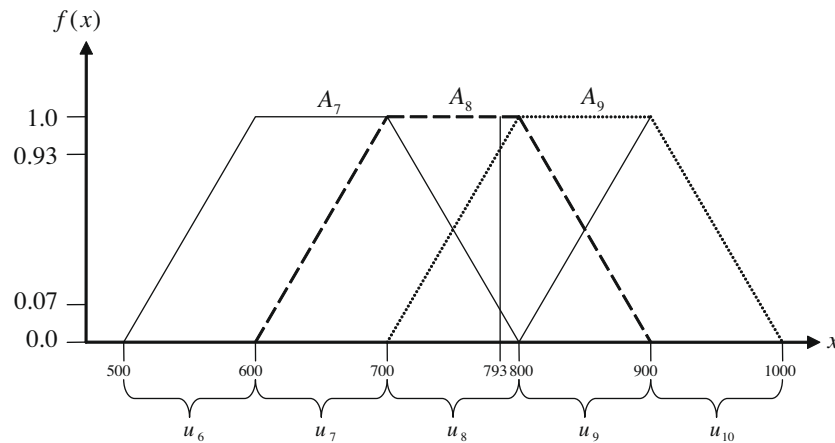


Fig. 2. Some examples of membership functions.

$V(2000/1/4, 2000/1/5)$
 $= (0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.07, 1.00, 0.93, 0.00, 0.00, 0.00).$

Table 1 lists the corresponding degrees of membership for all the differences.

Step 5. Neural Network Training

For the year 2000, there are 12 input and output nodes, respectively. The 12 input nodes are for the degrees of membership of $V(t-1, t)$, and the 12 output nodes are for the degrees of membership of $V(t, t+1)$. The number of hidden nodes is set to the sum of the number of input and output nodes, which is 24.

The degrees of membership for $d'(2000/1/4, 2000/1/5) = 793$ become the inputs of the neural network:

$V(2000/1/4, 2000/1/5)$
 $= (0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.07, 1.00, 0.93, 0.00, 0.00, 0.00).$

Those for $d'(2000/1/5, 2000/1/6) = 772$ are the outputs:

$V(2000/1/5, 2000/1/6)$
 $= (0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.28, 1.00, 0.72, 0.00, 0.00, 0.00).$

By repeating the process, we use all of the in-sample observations in Table 1 to train the neural network.

Step 6. Neural Network Forecasting

As an example, we use the degrees of membership for $d'(2000/10/2, 2000/10/3) = 554.48$ as the inputs, which are

$V(2000/10/2, 2000/10/3)$
 $= (0.00, 0.00, 0.00, 0.00, 0.46, 1.00, 0.54, 0.00, 0.00, 0.00, 0.00, 0.00).$

The outputs from the neural network are the forecasted degrees of membership for the next difference:

$V(2000/10/3, 2000/10/4)$
 $= (0.00, 0.03, 0.03, 0.09, 0.18, 0.30, 0.47, 0.38, 0.26, 0.11, 0.06, 0.03).$

Similarly, we can forecast all the degrees of membership for the other out-of-sample differences. Table 2 lists all the forecasted degrees of membership.

Step 7. Defuzzification

The forecasted difference between 10/3 and 10/4 is calculated as follows:

$$fd(2000/10/3, 2000/10/4) = \frac{0.00 \times 50 + 0.03 \times 150 + 0.03 \times 250 + 0.09 \times 350 + \dots + 0.03 \times 1150}{0.00 + 0.03 + 0.03 + 0.09 + 0.18 + 0.30 + 0.47 + 0.38 + 0.26 + 0.11 + 0.06 + 0.03} = 671.65.$$

Step 8. Forecasting

The defuzzified forecast is calculated as follows:

$$fd'(2000/10/3, 2000/10/4) = fd(2000/10/3, 2000/10/4) - 700 \\ = 671.65 - 700 = -28.35,$$

$$forecast(2000/10/4) = fd'(2000/10/3, 2000/10/4) \\ + obs(2000/10/3) \\ = -28.35 + 6143.44 = 6115.09.$$

The forecasts are depicted with the observations in Fig. 3.

Step 9. Performance Evaluation

For the year 2000, the RMSE for out-of-sample forecasting is 149.59.

6. Empirical analysis

6.1. Empirical results

To demonstrate the performance of the proposed model, we compare the out-of-sample RMSEs for different years with those in previous studies. We use Chen's model (Chen, 1996) to conduct similar forecasts as an example of a first-order model. Meanwhile, we also compare this model with a multivariate model (Huarng, Yu, et al., 2007) and a simple neural network model (Huarng & Yu, 2006b). Table 3 compares the forecasting results of the proposed model with those of the above-mentioned studies. The RMSEs of the proposed model are much smaller than their corresponding values in both the first-order model (Chen, 1996) and the multivariate model (Huarng, Yu, et al., 2007). Meanwhile, most RMSEs based on the proposed model are smaller than those of the counterpart neural network model (Huarng & Yu, 2006b). To sum up, the proposed model outperforms these models.

6.2. Discussions

The proposed model uses all the degrees of membership for neural network training and then for forecasting. In other words, $V(t-1, t) \rightarrow V(t, t+1)$. It means that more information has been

Table 1

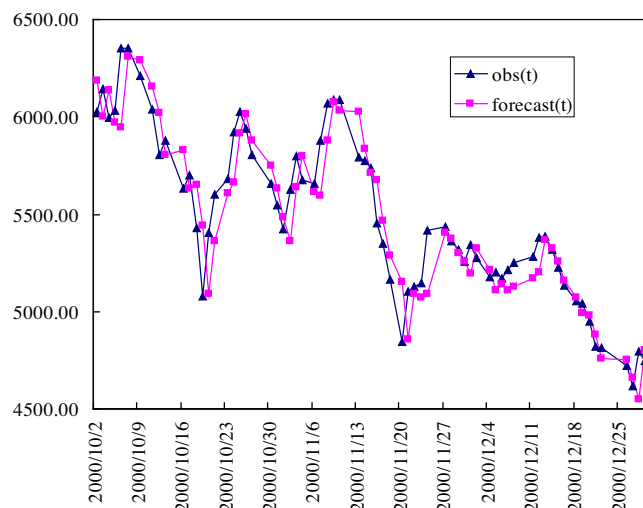
Degrees of membership for the differences in the year 2000.

$t-1$	$d'(t-1, t)$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}	A_{11}	A_{12}
2000/1/4	793	0	0	0	0	0	0	0.07	1	0.93	0	0	0
2000/1/5	772	0	0	0	0	0	0	0.28	1	0.72	0	0	0
2000/1/6	623	0	0	0	0	0	0.77	1	0.23	0	0	0	0
2000/1/7	957	0	0	0	0	0	0	0	0	0.43	1	0.57	0
2000/1/10	524	0	0	0	0	0.76	1	0.24	0	0	0	0	0
2000/1/11	918	0	0	0	0	0	0	0	0	0.82	1	0.18	0
2000/1/12	663	0	0	0	0	0	0.37	1	0.63	0	0	0	0
2000/1/13	616	0	0	0	0	0	0.84	1	0.16	0	0	0	0
2000/1/14	868	0	0	0	0	0	0	0	0.32	1	0.68	0	0
2000/1/15	824	0	0	0	0	0	0	0	0.76	1	0.24	0	0
...													
2000/12/19	608	0	0	0	0	0	0.92	1	0.08	0	0	0	0
2000/12/20	569	0	0	0	0	0.31	1	0.69	0	0	0	0	0
2000/12/21	694	0	0	0	0	0	0.06	1	0.94	0	0	0	0
2000/12/22	610	0	0	0	0	0	0.9	1	0.1	0	0	0	0
2000/12/26	593	0	0	0	0	0.07	1	0.93	0	0	0	0	0
2000/12/27	883	0	0	0	0	0	0	0	0.17	1	0.83	0	0
2000/12/28	647	0	0	0	0	0	0.53	1	0.47	0	0	0	0
2000/12/29	695	0	0	0	0	0	0.05	1	0.95	0	0	0	0

Table 2

Forecasted degrees of membership for the year 2000 (out-of-sample).

t	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}	A_{11}	A_{12}
2000/10/3	0.00	0.03	0.03	0.09	0.18	0.30	0.47	0.38	0.26	0.11	0.06	0.03
2000/10/4	0.01	0.03	0.03	0.07	0.17	0.40	0.55	0.33	0.20	0.14	0.06	0.03
2000/10/5	0.00	0.02	0.03	0.07	0.20	0.39	0.53	0.31	0.19	0.12	0.06	0.03
2000/10/6	0.00	0.01	0.01	0.02	0.07	0.34	0.71	0.45	0.16	0.09	0.01	0.01
2000/10/7	0.00	0.02	0.02	0.06	0.18	0.43	0.57	0.29	0.16	0.12	0.05	0.02
2000/10/9	0.01	0.03	0.03	0.06	0.17	0.41	0.56	0.33	0.20	0.14	0.06	0.03
2000/10/11	0.01	0.03	0.04	0.08	0.19	0.35	0.51	0.36	0.22	0.13	0.07	0.04
2000/10/12	0.01	0.03	0.04	0.10	0.18	0.25	0.43	0.42	0.26	0.10	0.07	0.04
2000/10/13	0.01	0.03	0.03	0.09	0.20	0.35	0.48	0.33	0.22	0.11	0.06	0.03
2000/10/14	0.01	0.03	0.04	0.10	0.18	0.24	0.43	0.43	0.27	0.10	0.07	0.04
...												
2000/12/19	0.00	0.02	0.02	0.05	0.18	0.44	0.58	0.29	0.17	0.13	0.04	0.02
2000/12/20	0.00	0.02	0.01	0.04	0.14	0.47	0.63	0.28	0.16	0.13	0.04	0.01
2000/12/21	0.00	0.02	0.02	0.05	0.16	0.43	0.58	0.31	0.19	0.14	0.05	0.02
2000/12/22	0.00	0.02	0.02	0.06	0.18	0.43	0.57	0.29	0.17	0.12	0.04	0.02
2000/12/26	0.00	0.02	0.01	0.04	0.14	0.47	0.63	0.28	0.16	0.13	0.04	0.01
2000/12/27	0.00	0.02	0.02	0.04	0.14	0.46	0.62	0.28	0.16	0.13	0.04	0.02
2000/12/28	0.00	0.02	0.03	0.07	0.13	0.28	0.52	0.48	0.36	0.12	0.05	0.03
2000/12/29	0.00	0.02	0.02	0.05	0.16	0.46	0.60	0.28	0.16	0.13	0.04	0.02

**Fig. 3.** The out-of-sample forecasts for the year 2000.

taken into consideration. Hence, the forecasting results are expected to improve. A similar study that also applies neural networks considered only $A_i \rightarrow A_j$ (Huang & Yu, 2006b). Only the most significant information was used for training and forecasting. The distinction is clear.

The proposed model can also forecast the relationships that never appeared in the in-sample observations. These empty relationships constitute an issue in relevant studies that applied only $A_i \rightarrow A_j$. Those studies needed to handle these issues using *ad-hoc* methods (Chen, 1996; Huang & Yu, 2006b), and the *ad-hoc* methods may have directly affected the forecasting results. Due to the coverage of all the degrees of membership, the proposed model can directly forecast these observations. So this issue can be resolved using the proposed model.

The drawback of taking all the degrees of membership for training and forecasting is that there can be too many fuzzy sets or inputs for the neural networks. Too many inputs may affect the performance of the neural networks. However, as the first step of the proposed model, we take the differences between observations. The purpose is to reduce the range of the universe of discourse.

Table 3

Performance evaluation by RMSEs.

	2000	2001	2002	2003	2004	Total
First-order model (Chen, 1996)	176.32	147.84	101.18	74.46	84.28	584.08
Multivariate model (Huarng, Yu, et al., 2007)	154.42	124.02	95.73	70.76	72.35	517.28
Neural network model (Huarng & Yu, 2006b)	152	130	84	56	N/A	N/A
This study	149.59	98.91	78.71	58.78	55.91	441.80

Then, we may have a smaller amount of intervals and fuzzy sets. Hence, the drawback can be amended.

The proposed model can easily be expanded into a multivariate fuzzy time series model in the future. Different advanced techniques can be applied to calculate the fuzzy relationships, too. Following the empirical results in this study, advanced techniques with the ability to handle nonlinearities can assist the fuzzy time series models in outperforming the original models.

7. Conclusions

This study proposes a fuzzy time series model that applies neural networks for training and forecasting. All the degrees of membership from the observations are taken into consideration. Due to the coverage of more information, the proposed model outperforms certain other fuzzy time series models based on the out-of-sample RMSEs. Another advantage of taking all the degrees of membership into consideration is that the proposed model can be applied to forecasting the observations that may not appear in the in-sample observations, which is a critical issue in some of the previous studies.

Attempting to cover all the degrees of membership may present a problem for neural networks, i.e., too many inputs. The proposed model takes the differences between observations at the very beginning, and so this problem can be solved smoothly. Once the number of inputs can be solved, the proposed model can easily be expanded to multivariate models.

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